Temporal Prediction of Weather

Preamble

The success of GaiaDhi's AgTech platform Holos is centred around increasing the accuracy of crop yield forecasting. Holos uses crop simulation models which base crop growth on crop science and inputs related to environment, and farming practices. The higher the accuracy of the inputs, the higher the accuracy of the models.

One of the highly dynamic, and hard to forecast environmental inputs is weather (key parameters: max temp, min temp, solar radiation, and rainfall), and they differ location to location, and season to season.

Project Details

Current Holos Weather Pipeline

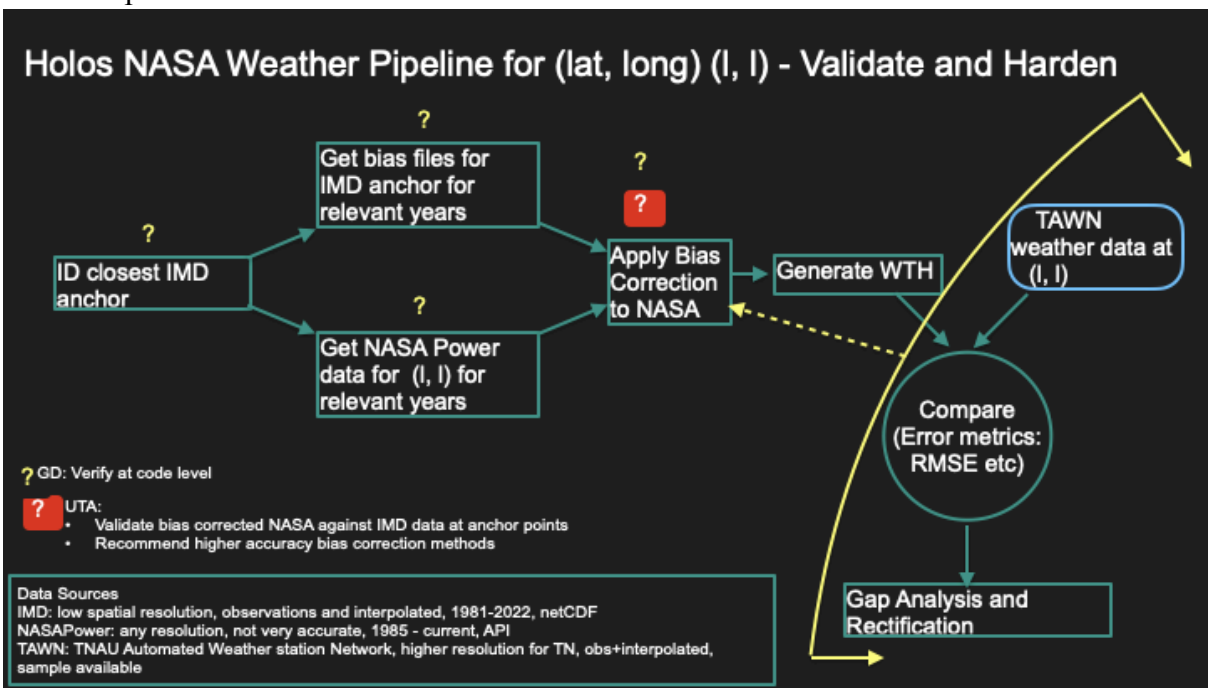
Holos uses NASA Power API to obtain current and historical weather at any latitude and longitude. Historical weather data as part of crop yield forecast for the upcoming season. While NASA Power is very flexible, we have been informed it is not localized and may also vary seasonally with localized observation.

IMD Resolution	
Spatial - coarse	Tmin and Tmax at 1 deg (111 km x 111 km), Rain at 0.25 deg (28km x 28km), no SRAD
Temporal	1951 - 2020
NASA POWER Resolution	
Spatial - fine	any, incl SRAD
Temporal	1985 - current

In an attempt to localise (ie., increased spatial prediction), Holos does the following

1. IMD has provided low resolution observed (ground weather stations and interpolated). The assumption is that this data is ground truth. IMD does not provide solar radiation.
2. Holos has a precomputed database of deltas between NASA Power data and IMD (delta = IMD - NASA POWER), for points (lat, lon) where available. Call these "anchor points". The deltas are called "bias"es.
3. At time of simulation: for a point of simulation (PoS), which may vary from an anchor point, a bias correction is applied:
    1. NASA Power for that point is obtained
    2. Anchor point closest to simulation point is determined, and the delta for that point is extracted

3. Weather for that point = NASA POWER weather for that point + delta. For a point where IMD data is available, this bias correction would end up at the IMD value
4. By bias correcting NASA Power to observed and interpolated IMD localised data, we expect to have localised NASA Power.



Prior UTA Capstone Projects

Goal:

Replace the bias correction functionality with an ML model for spatial (not temporal) prediction. Seasonality should be taken into consideration.

Spring 2024

Git Repo: https://github.com/gaiadhi/Prototypes/tree/main/Weather (at root)

LR was identified as an ideal model for higher resolution spatial prediction of min and max temps

Presentation:
https://github.com/gaiadhi/Prototypes/blob/main/Weather/Gaiadhi_ProjectPresentation.pptx

Fall 2024

Git Repo: https://github.com/gaiadhi/Prototypes/tree/main/Weather/UTA_Team7_Project

Random Forest was identified as an ideal model for higher resolution spatial prediction of rainfall/precipitation.

Presentation:
https://github.com/gaiadhi/Prototypes/blob/main/Weather/UTA_Team7_Project/V.4%20Project%20framework%20group%207.pdf

High Level Problem Statement

Goal:

Temporal prediction of key weather params (max temp, min temp, rain, solar radiation.

Possible approaches:

1. Evaluate models identified in prior UTA Capstone projects for temporal prediction
2. Evaluate Process based ML models such as GraphCast
    a. areas of accuracy may need to be understood - USA, India)
    b. Can trained models be used for purely inferring or do we need to train?
    c. What is the compute infra needed to train and/or inference? Open infras?
3. Evaluate temporal prediction efficacy for 7 days and then to 14 days max

Note of GrapCast

Google's GraphCast is based on outputs from ECMWF's physical models.

Pros:

1. ECMWF is a well recognized weather model
2. GraphCast claims to be more accurate than ECMWF, but is based on results fromECMWF
3. If global, use it as a global bias source, rather than just IMD for India and another local source for another country?

Cons:

1. Same coarse resolution as IMD
2. Does this need to be localised? This will assess the global bias source applicability.